

A trivial-line counterexample to the symmetric-power positivity question

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Status

Full counterexample to the second question in Remark 5.9 as written. The counterexample does not address Griffiths’ ample-bundle conjecture or the Banach-bundle Koszul–Malgrange problem elsewhere in the source.

Question and convention

The source is D. Beltiță and J. E. Galé, *Reproducing kernels and positivity of vector bundles in infinite dimensions*, arXiv:1402.0458. Remark 5.9 on PDF page 17 asks whether a holomorphic vector bundle must be Griffiths positive whenever all sufficiently high symmetric powers are globally generated.

see for instance [De12] Ch. V, Cor. 15.6], where this line bundle was denoted by $\mathcal{O}(-1)$.

Remark 5.9. In connection with Theorem 5.4 we recall that there exist several open problems on sufficient conditions for Griffiths positivity. There is for instance Griffiths’ problem ([Gr69]) which asks whether or not every ample holomorphic vector bundle with compact base is Griffiths positive.

It is also unknown whether any holomorphic vector bundle is Griffiths positive if there exists some integer $k_0 \geq 1$ such that the symmetric k th tensor powers of that bundle are globally generated for all $k \geq k_0$. See however [De12] Ch. VII, Cor. 11.6] for an affirmative answer to that problem in the case of the line bundles. From the perspective of the above Theorem 5.4 the problem would be to construct reproducing kernels on some Hermitian vector bundle by using some reproducing kernels on symmetric tensor powers of that bundle.

This wording is strict. In Definition 4.5 on PDF page 14, Griffiths nonnegativity means that the associated Hermitian curvature form $\Psi_z(v, v)$ is positive semidefinite. “Griffiths positive” additionally requires $\Psi_z(v, v) \neq 0$ for every nonzero base tangent vector v .

We now introduce the notion of Griffiths positivity of bundle-valued differential 2-forms, which goes back to [Gr69]; see also [GH78], [Lz04], [De12]. In the case of infinite-rank vector bundles, a version of this notion was also used in [Ber09].

Definition 4.5. Let $\Pi: D \rightarrow Z$ be any Hermitian holomorphic bundle. A bundle-valued differential form $\omega \in \Omega^2(Z, \text{End}(\Pi))$ is *Griffiths nonnegative* if for every $z \in Z$ the bounded $\mathbb{R}$ -bilinear map $\omega_z: T_z Z \times T_z Z \rightarrow \mathcal{B}(D_z)$ satisfies the conditions $\omega_z \in \text{Skew}(T_z Z, \mathcal{B}(D_z))$ and $\Psi_z(x, x) \geq 0$ in $\mathcal{B}(D_z)$ for all $x \in T_z Z$, where $\Psi_z \in \text{Herm}(T_z Z, \mathcal{B}(D_z))$ is the sesquilinear map which canonically corresponds to ω_z via Remark 4.4. If moreover for every $x \in T_z Z \setminus \{0\}$ we have $\Psi_z(x, x) \neq 0$, then we say that ω is *Griffiths positive*.

In the case of the bundles with a finite-dimensional base, the following result can

For a line bundle, positive semidefinite and nonzero is strictly positive.

Counterexample

Proposition. *Let X be any compact connected Riemann surface and let $L = \mathcal{O}_X$ be its trivial holomorphic line bundle. Then every symmetric power $S^k L$ is globally generated, but L admits no Griffiths-positive Hermitian metric.*

Proof. For every $k \geq 1$,

$$S^k L = L^{\otimes k} \simeq \mathcal{O}_X.$$

The constant holomorphic section 1 is nonzero at every point, so it generates every fiber. Thus the hypothesis in Remark 5.9 holds with $k_0 = 1$.

Suppose that a Hermitian metric h on L had Griffiths-positive Chern curvature. Since L is a line bundle and X is a curve, its normalized Chern curvature would be a strictly positive real 2-form representing $c_1(L)$. Hence

$$\int_X c_1(L, h) > 0.$$

But L is topologically trivial, so $c_1(L) = 0$ and Chern–Weil theory gives

$$\int_X c_1(L, h) = \langle c_1(L), [X] \rangle = 0,$$

a contradiction.

Equivalently, in the global holomorphic frame 1 one can write $h(1, 1) = e^\varphi$ for a smooth real function φ on X . The curvature is, up to the source’s fixed sign convention, $\partial\bar{\partial}\varphi$. Its integral is zero by Stokes, whereas strict Griffiths positivity would make the corresponding real form have strictly positive integral. $\square$

The same obstruction gives a higher-rank family: the trivial bundle $\mathcal{O}_X^{\oplus r}$ has every symmetric power globally generated. If its curvature were Griffiths positive in Definition 4.5, then the trace curvature would be strictly positive on every nonzero tangent direction, while it represents $c_1(\det \mathcal{O}_X^{\oplus r}) = 0$.

Interpretation

The source itself distinguishes nonnegative from positive curvature and proves in Theorem 4.6 that a globally generated bundle has a Griffiths- *nonnegative* metric. The trivial bundle is consistent with that theorem: its flat metric has zero curvature. The line-bundle statement cited in Remark 5.9 is therefore valid only with semipositivity/nonnegativity (or with a stronger hypothesis such as ampleness that excludes the trivial bundle), not with the strict Definition 4.5 used in the question.

References

- [1] D. Beltiță and J. E. Galé, *Reproducing kernels and positivity of vector bundles in infinite dimensions*, arXiv:1402.0458.
- [2] J.-P. Demailly, *Complex Analytic and Differential Geometry*, Chapter VII, §11.